Lab 6.1 – Configuring EtherChannel

Important for Online Workshop Students Only:

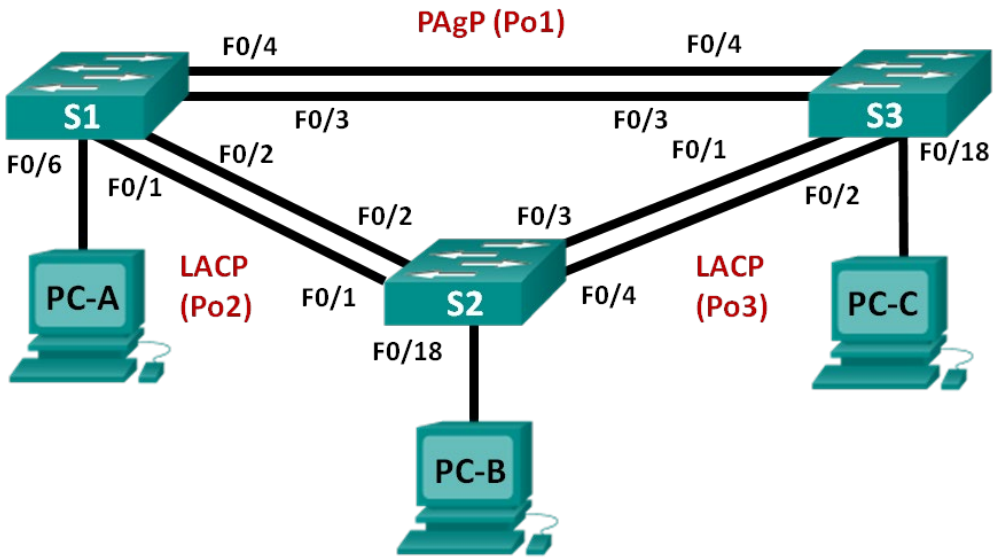
You must use Packet Tracer v8.2.2 or higher to complete this lab.

Please rename your Packet Tracer file to FirstName_LastName_sNumber_Lab6.1.pka

Scoring

No.		Marks
1	PCs	12
2	Switches	191

Topology



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Addressing Table

Device	Interface	IP Address	Subnet Mask
S1	VLAN 99	192.168.99.11	255.255.255.0
S2	VLAN 99	192.168.99.12	255.255.255.0
S3	VLAN 99	192.168.99.13	255.255.255.0
PC-A	NIC	192.168.10.1	255.255.255.0
PC-B	NIC	192.168.10.2	255.255.255.0
PC-C	NIC	192.168.10.3	255.255.255.0

VLAN Table

VLAN	Name	Interface Assigned
10	Staff	S1: F0/6 S2: F0/18 S3: F0/18
99	Management	VLAN 99
555	Parking_Lot	S1: F0/5, F0/7-24, G0/1-2 S2: F0/5-17, F0/19-24, G0/1-2 S3: F0/5-17, F0/19-24, G0/1-2
1000	Native	N/A

Objectives

Part 1: Configure Basic Switch Settings

Part 2: Create VLANs and Assign Switch Ports

Part 3: Configure PAgP

Part 4: Configure LACP

Background / Scenario

Link aggregation allows the creation of logical links that are comprised of two or more physical links. This provides increased throughput beyond using only one physical link. Link aggregation also provides redundancy if one of the links fails.

In this lab, you will configure EtherChannel, a form of link aggregation used in switched networks. You will configure EtherChannel using Port Aggregation Protocol (PAgP) and Link Aggregation Control Protocol (LACP).

Note: PAgP is a Cisco-proprietary protocol that you can only run on Cisco switches and on switches that are licensed vendors to support PAgP. LACP is a link aggregation protocol that is defined by IEEE 802.3ad, and it is not associated with any specific vendor.

LACP allows Cisco switches to manage Ethernet channels between switches that conform to the 802.3ad protocol. You can configure up to 16 ports to form a channel. Eight of the ports are in active mode and the

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other eight are in standby mode. When any of the active ports fail, a standby port becomes active. Standby mode works only for LACP, not for PAgP.

Note: For students in on-campus workshops only: Make sure that the switch has been erased and has no startup configurations. Please refer to Appendix A

Note: For students in online workshops only: In Packet Tracer, you must use the 2960 switch

Part 1: Configure Basic Switch Settings

In Part 1, you will set up the network topology and configure basic settings on the PC hosts and switches.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: On-campus students: Initialize and reload the switches (See Appendix A)

Step 3: Configure the PCs.

Assign IP addresses to the PCs according to the Addressing Table.

Step 4: Configure basic settings for each switch.

a. Assign a device name to the switch.

b. Disable DNS lookup.

c. Assign **class** as the privileged EXEC encrypted password.

d. Assign **cisco** as the console password and enable login.

e. Assign **cisco** as the vty password and enable login.

f. Configure logging synchronous for the console line and vty line.

g. Encrypt the plaintext passwords.

h. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
```
S1(config)# banner motd 'Unauthorised access is strictly prohibited and you
will be prosecuted fully.'
```

i. Configure the switch for SSH access. Use **1024 bit rsa** encryption, domain-name **networklab.com**, username **admin,** privilege level **15,** and password **cisco**. Update the vty lines to use local login and ssh.

j. Shut down all switchports except the ports connected to the PCs. Remember to include the Gigabitethernet ports.

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Part 2: Create VLANs and Assign Switch Ports

In Part 2, you will create VLANs as specified in the table above on all switches. You will then assign the VLANs to the appropriate interface and verify your configuration settings. Complete the following tasks on each switch.

Step 1: Create VLANs on all the switches.

a. On all the switches, create and name the required VLANs from the VLAN Table above.

b. Configure and activate the management interface on each switch using the IP address information in the Addressing Table.

c. Assign all unused ports on the switch to the Parking_Lot VLAN, configure them for static access mode, and maker sure they are administratively deactivated (shutdown). Review Lab 5 for switchport access configuration. Note: Only apply this to unused ports. Ports between PCs and between switches are in use and should not be configured in this step.

   Example for S1:

```
S1(config)# interface f0/5
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 555
S1(config-if)# shutdown
S1(config-if)# exit
S1(config)# interface range f0/7-24
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 555
S1(config-if)# shutdown
S1(config-if)# exit
S1(config)# interface range g0/1-2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 555
S1(config-if)# shutdown
S1(config-if)# exit
```

Step 2: Assign VLANs to the correct switch interfaces.

a. Assign switch ports with attached hosts to the appropriate VLAN (specified in the VLAN table above) and configure them for static access mode. Note that you have already configured the Management VLAN by adding an IP address to VLAN 99 on each switch earlier in this configuration.

   Example for S1:

```
S1(config)# interface f0/6
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# exit
```

b. Issue the **show vlan brief** command and verify that the VLANs are assigned to the correct ports.

Part 3: Configure PAgP

PAgP is a Cisco proprietary protocol for link aggregation. In Part 3, a link between S1 and S3 will be configured using PAgP.

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Step 1: Configure PAgP on S1 and S3.

For a link between S1 and S3, configure the f0/3and f0/4 ports on S1 with PAgP desirable mode and the f0/3 and f0/4 ports on S3 with PAgP auto mode.

The **mode desirable** option enables the switch S1 to actively negotiate to form a PAgP link, while the **mode auto** option indicates that the switch S3 is passively waiting for the other side to initiate negotiations.

For the **channel-group** use the **port channel number 1** for the link between S1 and S3.

Enable the ports after PAgP modes have been configured (no shutdown).

Example for S1:

```
S1(config)# interface f0/3
S1(config-if)# channel-group 1 mode desirable
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# interface f0/4
S1(config-if)# channel-group 1 mode desirable
S1(config-if)# no shutdown
S1(config-if)# exit
```

Example for S3:

```
S3(config)# interface f0/3
S3(config-if)# channel-group 1 mode auto
S3(config-if)# no shutdown
S3(config-if)# exit
S3(config)# interface f0/4
S3(config-if)# channel-group 1 mode auto
S3(config-if)# no shutdown
S3(config-if)# exit
```

Step 2: Examine the configuration on the ports.

Currently the F0/3, F0/4, and Po1 (Port-channel1) interfaces on both S1 and S3 are in access operational mode with the administrative mode in dynamic auto. Verify the configuration using the **show run** and **show interfaces** *interface-id* **switchport** commands, respectively. The example configuration outputs for F0/3 on S1 are as follows:

```
S1# show run
Building configuration...

Current configuration : 103 bytes
!
interface FastEthernet0/3
 channel-group 1 mode desirable

S1# show interfaces f0/3 switchport
Name: Fa0/3
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
```

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```
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk Native VLAN tagging: enabled
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk associations: none
Administrative private-vlan trunk mappings: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL

Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

Step 3: Verify that the ports have been aggregated.

```
S1# show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         PAgP      Fa0/3(P)    Fa0/4(P)

S3# show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
```

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```
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         PAgP      Fa0/3(P)    Fa0/4(P)
```

What do the flags, SU and P, indicate in the Ethernet summary?

_____

_____

Step 4: Configure trunk ports.

After the ports have been aggregated, commands applied at the port channel interface (Po1) affect all the links (ports) that were bundled together. This means that if you put configuration on Po1, that configuration will be automatically applied to the ports that make up the Po1 channel (in this case f0/3 and f0/4).

Manually configure the Po1 ports on S1 and S3 as trunk ports and assign them to native VLAN 1000. Specify that VLANs 10, 99, and 1000 are allowed to cross the trunk. Note that VLAN 555 is not allowed to cross the trunk.

Example for S1 (note configuring two interfaces at once):

```
S1(config)# interface po1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 1000
S1(config-if)# switchport trunk allowed vlan 10,99,1000
S1(config-if)# exit
```

Verify that the ports are configured as trunk ports.

a. Issue the **show run** commands on S1 and S3. What commands are listed for F0/3 and F0/4 on both switches S1 and S3? Compare the results to the running configuration for the Po1 interface? Record your observation.

_____

_____

b. Issue the **show interfaces trunk** and **show spanning-tree** commands on S1 and S3. What trunk port is listed? What is the native VLAN?

_____

_____

From the **show spanning-tree** output, what is the port cost and port priority for the aggregated link?

_____

_____

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Part 4: Configure LACP

LACP is an open source protocol for link aggregation developed by the IEEE. In Part 4, the link between S1 and S2, and the link between S2 and S3 will be configured using LACP.

Step 1: Configure LACP between S1 and S2

For the link between S1 and S2, on S1 we cannot use port channel number 1 because we had already used number 1 in the previous Part 3 to connect S1 and S3 using interfaces f0/3 and f0/4. So,we use the **port channel number 2** for the link between S1 and S2. Configure the ports on S1 with LACP active mode and the ports on S2 with LACP passive mode. Active mode indicates that the switch S1 actively tries to negotiate that link as LACP, while passive mode indicates that the switch S2 is passively waiting for the other side to initiate negotiations. Enable the ports after LACP modes have been configured.

Example for S1

```
S1(config)# interface f0/1
S1(config-if)# channel-group 2 mode active
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# interface f0/2
S1(config-if)# channel-group 2 mode Active
S1(config-if)# no shutdown
S1(config-if)# exit
```

Example for S2:

```
S3(config)# interface f0/1
S3(config-if)# channel-group 2 mode passive
S3(config-if)# no shutdown
S3(config-if)# exit
S3(config)# interface f0/2
S3(config-if)# channel-group 2 mode passive
S3(config-if)# no shutdown
S3(config-if)# exit
```

Step 2: Configure trunk ports

After the ports have been aggregated, commands applied at the port channel interface affect all the links that were bundled together. Manually configure the Po2 ports on S1 and S2 as trunk ports and assign them to native VLAN 1000. Specify that VLANs 10, 99, and 1000 are allowed to cross the trunk (See step 4 in the previous section for the commands).

Step 3: Verify that the ports have been aggregated.

What protocol is Po2 using for link aggregation? Which ports are aggregated to form Po2? Record the command used to verify.

_____

Step 4: Configure LACP between S2 and S3.

a. Configure the link between S2 and S3 as **Po3** and use LACP as the link aggregation protocol. Configure the ports on S2 with LACP **active mode** and the ports on S3 with LACP **passive mode**. Enable the ports after LACP modes have been configured.

b. After the ports have been aggregated, commands applied at the port channel interface affect all the links that were bundled together. Manually configure the Po3 ports on S2 and S3 as trunk ports and assign them to native VLAN 1000. Specify that VLANs 10, 99, and 1000 are allowed to cross the trunk.

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c. Verify that the EtherChannel has formed.

Step 5: Verify end-to-end connectivity.

Verify that all devices can ping each other within the same VLAN. If not, troubleshoot until there is end-to-end connectivity.

Can PC-A ping PC-B? _____

Can PC-A ping PC-C? _____

Can PC-B ping PC-C? _____

Can S1 ping S2? _____

Can S1 ping S3? _____

Can S2 ping S3? _____

Reflection Question

What could prevent EtherChannels from forming? Please give 5 possible reasons.

_____

_____

_____

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Appendix A: Initializing and Reloading a Router and Switch

Step 1: Initialize and reload the router.

a. Console into the router and enable privileged EXEC mode.

```
Router> enable
Router#
```

b. Type the **erase startup-config** command to remove the startup configuration from NVRAM.

```
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
Router#
```

c. Issue the **reload** command to remove an old configuration from memory. When prompted to **Proceed with reload**, press Enter to confirm the reload. (Pressing any other key aborts the reload.)

```
Router# reload
Proceed with reload? [confirm]
*Nov 29 18:28:09.923: %SYS-5-RELOAD: Reload requested by console. Reload Reason:
Reload Command.
```

Note: You may be prompted to save the running configuration prior to reloading the router. Type **no** and press Enter.

```
System configuration has been modified. Save? [yes/no]: no
```

d. After the router reloads, you are prompted to enter the initial configuration dialog. Enter **no** and press Enter.

```
Would you like to enter the initial configuration dialog? [yes/no]: no
```

e. You are prompted to terminate autoinstall. Type **yes** and then press Enter.

```
Would you like to terminate autoinstall? [yes]: yes
```

Step 2: Initialize and reload the switch.

a. Console into the switch and enter privileged EXEC mode.

```
Switch> enable
Switch#
```

b. Use the **show flash** command to determine if any VLANs have been created on the switch.

```
Switch# show flash
Directory of flash:/

    2  -rwx        1919   Mar 1 1993 00:06:33 +00:00  private-config.text
    3  -rwx        1632   Mar 1 1993 00:06:33 +00:00  config.text
    4  -rwx       13336   Mar 1 1993 00:06:33 +00:00  multiple-fs
    5  -rwx    11607161   Mar 1 1993 02:37:06 +00:00  c2960-lanbasek9-mz.150-2.SE.bin
    6  -rwx         616   Mar 1 1993 00:07:13 +00:00  vlan.dat

32514048 bytes total (20886528 bytes free)
Switch#
```

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c. If the **vlan.dat** file was found in flash, then delete this file.

```
Switch# delete vlan.dat
Delete filename [vlan.dat]?
```

d. You are prompted to verify the filename. At this point, you can change the filename or just press Enter if you have entered the name correctly.

e. You are prompted to confirm deleting this file. Press Enter to confirm deletion. (Pressing any other key aborts the deletion.)

```
Delete flash:/vlan.dat? [confirm]
Switch#
```

f. Use the **erase startup-config** command to erase the startup configuration file from NVRAM. You are prompted to confirm removing the configuration file. Press Enter to confirm to erase this file. (Pressing any other key aborts the operation.)

```
Switch# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
Switch#
```

g. Reload the switch to remove any old configuration information from memory. You are prompted to confirm reloading the switch. Press Enter to proceed with the reload. (Pressing any other key aborts the reload.)

```
Switch# reload
Proceed with reload? [confirm]
```

Note: You may be prompted to save the running configuration prior to reloading the switch. Type **no** and press Enter.

```
System configuration has been modified. Save? [yes/no]: no
```

h. After the switch reloads, you should be prompted to enter the initial configuration dialog. Type **no** and press Enter.

```
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>
```

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